

ActInf Textbook Group ~ Cohort 1 ~ Meeting 9

(Chapter 4 pt. 2)

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of these questions i think we talked about the oh this is chapter two uh let's go to chapter four okay so we talked about belief policy state i'm not sure if we talked about this question [Music] in the discussion of active inference in pomdp belief updating about policies we find that the posterior that minimizes the free energy does it posterior at time t oh we did discuss this yeah we did discuss this last time so uh this question i think um we did not discuss π being a policy or a model last time or did we i think we didn't get to this yeah i don't think we discussed that all right so we can open it up and um we'll start here i guess with the the most upvoted so the question reads what is π on page 69 the authors write at each time step the current state is conditionally dependent on the state at the previous time and on the policy π currently being perceived pursued then on page 71 they write thus we can interpret the priors of equation 4.6 combined with the likelihood of equation 4.5 as expressing a model π of a behavioral sequence so which is it policy or model and then they suggest a rewrite of the sentence they say that thus we can 4.5 and the transition probabilities $b_{\tau|\pi}$ as expressing a model for a behavioral sequence where the model is a function of policy π and then there's some discourse here so in this reframing can we say that the model simulates the agent in the environment as as if it had taken the actions in the policy um and i don't know does anyone have any i don't know who asked that question um oh eric that was you that was your question uh and then does anyone want to maybe take a stab at answering that question or we can continue reading what was written here uh so what's written here says that on page 69 the authors also say policies here may be thought of as indexing alternative trajectories or sequences of actions that could be followed on page 71 they defined the likelihood in equation 4.5 as a matrix a that expresses the probability of an outcome they describe the priors of equation 4.6 as the prior over the initial state vector d and beliefs about how the state at one time transitions to the state at the next time matrix b and they also say that the transitions are conditionally dependent on the policy chosen because of this conditional dependence we can

see how the policy influences the model and why the authors use may use these the terms interchangeably does anyone have any comments there um i guess i would say that that um answer in discourse kind of um agrees with my that i proposed which is that strictly speaking um we should treat the pie as policy but it behaves as a model but once it's executed it's implemented then the model become the model has become a function of the policy that's been we've seen so far so i i'd say that's a you know that's compatible um interpretation so i think strictly speaking the um it would be better if the text was consistent and called π a policy and then um say yeah the model is a function of path that'd be my interpretation yeah i definitely agree with that i think that the model is policy specific so so yeah but they could do a better job of i mean the variables as we've seen are so ambiguous anyway so it could definitely be a lot better uh and by the way uh about the topic of consistency um actually i gave some thoughts about the issue we were discussing in yesterday's math learning session and i also consulted some other papers and active inference and in all of them the notation for matrices and vectors is used consistently as in almost every linear algebra textbook so i'm seriously beginning to suspect that every instance of a matrix or a vector not written in boldface is a typo either in chapters or appendices because otherwise i really cannot find any justification behind using two different types of notation so i'm going to use this assumption as my prior for the rest of the book unless someone has any belief updating observation she wants to share so just to summarize for for the people that weren't there yesterday we had a big debate about what does the bold um typeface signify and it wasn't so much a debate but but just like a mutual trying to come to a mutual understanding so it's this question here does bold lettering mean this is a matrix but seemingly also many non-bold letters are matrix or does bold lettering mean sufficient statistics of um as as on these page examples and it really was uh yeah we couldn't come to a single unified answer there because there are many instances where it seems like it should be a matrix but it's not bold and there are many instances where um sufficient statistics like are used but it's also not bold so so yeah we couldn't really conclude that um but i think ali that that's a good assumption there uh so we also talked about this a little bit last time but we didn't really get fully into this question what is the use of categorical distributions in equation 4.5 the second line is supposed to explain the cat notation but i have problems to understand the advantage of the notation could someone explain it in simple words does anyone want to try to explain that i can pull up the equation uh well i think every regular matrix and vector is by definition an array of ordered elements uh but in a categorical distribution uh we don't necessarily assign any any specific order to the elements

so uh the reason behind using the categorical categorical distribution might be to just jettison any uh orderliness in the elements and treat them as just uh an on an unordered set of the probabilities i wrote an answer to this one if you want to pull that out so you can just write that so i hope this is helpful it's basically the way i see it is you have the math the math notation there but when you need to actually implement that you need to carry it out then you need to turn that into operations you can perform and so the um when you call when you say that a is this categorical object that means that you you're turning it into an actual matrix with elements and so the ordering is important actually because then the ordering of the elements in the matrix says how that matrix is going to apply to um the objects it applies to which are the um the belief state s and then the your observation o so um basically it says the way you carry out inferring what o will be the probability of o given a belief state is you do a matrix multiply and the because our in in this you know in this example the the um states are not continuous they're categorical then you need a categorical matrix you know elements in order to do map from the categories of belief states to the categories of observation states and those are all those are both distributions that gets transformed by the thanks eric that was super helpful does anybody have any further question or comment on this one uh sorry eric but isn't the categorical notation special case of multinomial distribution categorical distribution special cases of multinomial distribution i don't know that yeah because that was my understanding that the main difference between uh any categorical distribution and uh well i mean uh the the lack of for instance if we of course you're right that in any we necessarily have some ordered elements in order uh for us to be able to compute the math on those matrices uh but on the other hand uh for categorical distributions at least that was my understanding that uh we don't necessarily treat them as ordered elements such as i don't know the ordered vectors or ordered matrices as the normal matrix because uh if that was the case we could just uh designate this this distribution as just a regular matrix right so uh i don't get what's the reason behind using uh the notation here a cat stephen can you mute you're super loud over there sorry yeah i i guess so i mean yeah so the mathematical notation doesn't have an ordering it just says distribution is related to another distribution through this this this relation and it's when you get to the mechanics of how do you how do you do that that's where you care that you have the matrix with with the vector ordering so yeah i don't know what a actually means otherwise other than it's just describing it it's saying i don't think it's like an operator i think it's just saying what it is that makes sense steven if you're trying to talk we can't hear i feel like steven is trying to share an idea with us but is unable to communicate effectively through the

technological affordances he's been given at the time like the gain on your mic is way up or something uh so it's picking up more background noise than you so just looking at this equation let's try that again so just looking at this equation can you guys hear me okay yeah yeah okay the second part it's like the it's like an ordered matrix a_{ij} where the probability of the observations are equal to index i and the states equal to index j something like that i don't know some it looks like it is ordered yeah the matrix itself is ordered yeah and so the second line is is actually by how the authors are defining this cat a so so cat a represents this ordered matrix a_{ij} that's at least what it what it says in the textbook so it says the likelihood expressing the probability of observations conditioned on time τ given states s conditioned on time τ is equal to the categorical distribution a_{ij} if i may sorry sorry yaku go ahead sorry my internet is a bit is lagging a bit so there might be a delay and i might break off but i just want to comment that i think that the cat notation is really just to uh like reinforce the the notion that it's in discrete time but i think it's kind of redundant in this case because since we know that we are in discrete time we could have just written uh a bold to uh signify that it's a matrix and when it's a matrix it's probably not going to be a continuous distribution anyway so i think bolt a and bolt a is pretty much interchangeable so i think that some of the new math um is using some of the affordances from category theory more like some of the the newer things and i think like the forward direction that carl and thomas and some of the people that really helped derive and advance some of this math i think that they're going to be leveraging category theory methods like the renormalization group and the pullback attractor um more in the future and so that might be also like part of the reason that they are moving toward this categorical notation so just a little maybe foreshadowing there too and may i just make some one additional note about this categorical distribution well the thing that confused me about uh this this concept of categorical distribution is uh the wikipedia's definition of it because here it says there is no innate underlying ordering of these outcomes but numerical labels are often attached for convenience in describing the distribution so that was my confusion um i don't think it would be the first time that we've seen a traditional mathematical notation slightly and somewhat abused in active inference probably also won't be the last but definitely i think it's a point that we can put forward for to the authors for additional clarification if there's no further anybody has anything else to say about the categorical distributions here if not we can move on to maybe the next so this next question reads in message passing is there a decay of information as the distance between the variable x and individual markov blanket constituents increases is implementation of information decay in message passing an option for model

implementation um i think that this is a super interesting question and it's something that i've thought about before a lot but but i think um yeah let me open it up to anyone and see if they want to address it before we can get into some of the discourse well doesn't that all all depend upon the the chain of loss between one stage of the message passing in another and if you have no loss then you don't have any decay but you know the more stages you have where you get a little bit of diffusion or loss then you're going to have decay just a function of the parameters well so our message is passed within the same markov blanket or between things that are partitioned with their own markov blanket that's another uh i also agree with eric as i mentioned yesterday uh i also believe that uh well this message this information loss is not accounted for at least inherently in active inference formalism uh because it depends on the specific situations those information propagate because otherwise uh we couldn't have uh any uh general or all-encompassing formalism for accounting for these mess information laws so we see in this box they talk about markov blankets and the message passing so here it's variational message passing this involves messages from all constituents of the markov blanket of x including the parents via the conditional probability of x given its parents and the children the latter depends on the conditional probability of the children of x given all of their parents which include x note the expectation includes the children and parents of the children as parents of the children x we divide by q of x to ensure the expectation includes the blanket only so i think the messages are like all of the things inside of the blanket send a message to something else in another blanket at least that's how i interpret um this way of active inference and so it would make sense that um like from within the blanket does everyone have the same is there like a high degree of mutual information shared in the markov blanket i think like conditional dependence is what what um you know qualifies as as like conditional independence is what um establishes the markov blanket so within constituents under the blanket do they all have all of the same information all of the time that's something that's kind of confusing to me you know one thing i'd point out about figure 4.3 so in the previous chapters um you know we have this this picture of there being the world and your model of the world and and that and those being and your model of the world a markov blanket itself and it interfaces with the world through the the um observation variables and the act and the action variables this picture doesn't have that breakdown so it's got state there but it doesn't say what is that is that state all internal state is it external state or a mixture of them and it's apparently it's mixture because you've got policy that can affect state which means that you're going to have action is you know is going to be part of the state transition as well as and then you have

the observations coming out at the bottom um but i think if we're going to talk about markov blankets then i would like to see this picture be exploded into the internal you know the state of the machine versus the state of of the external world and how those inter interact with one another because that's what the markov blanket does is it compartmentalizes the information well and and so that is the the pomdp uh model and figure 4.3 um and this really figure 4.4 um is like makes me scratch my head quite a bit this is the image that is going to um like represent message passing in a bayesian framework and so um the caption says on the right dependencies between different variables in the belief updating scheme outlined in the main text intuitively current beliefs about states under each policy at each time are compared with those that would be predicted given beliefs about states at other times this is one and current outcomes to calculate prediction errors these errors then drive updating beliefs this is two given beliefs about states under each policy we can then calculate the gradients of the expected free energy this is three these are combined with the outcomes predicted under each policy omitted from the figure to compute beliefs about policies this is for and then using a bayesian model average we can then compute posterior beliefs about states averaged over policies this is five uh and this kind of leads us into the next question but like i i'm not able to really interpret this figure 4.4 in any better way so if anybody has like a plain english description of um how this works i would love to hear it we can also um we can talk a little bit about um the a question that was raised about this figure and also about this like squiggle sigma uh because there's a great like i don't know i mean we kind of had a hard time figuring out the equation and the squiggle sigma and what that actually stands for so we could move to that discussion because i think it's maybe related to this message passing discussion but let's read the discourse here first um on the message passing question in the discourse what is the mechanism for signal decay so this is um information decay implementation of information decay is the message a thing behaving as an active inference agent itself or a special piece of information not behaving as an active inference system itself is there a blanket impedance mismatch i think brock contributed this i'm not sure if he's here today um and then message passing is described through active inference usually is a hierarchy not a lateral transfer like within constituents in the blanket so the key thing here is how you define the markup blanket either there's a partition between particles cells or whatever or there's no partition in blanket and everything under the blanket is conditionally dependent does conditional dependence preclude message passing are message passed within nodes in the same blanket i have not seen this and would love if someone pointed me to some references so that's that was

me i wrote the last part um because i haven't seen message passing under the same markov blanket it's always been like from one markov blanket partitioned object to a different markov blanket partitioned object but that moves us into um if anybody has any comments feel free well i would say message passing is used to do the inference so it's within a markov blanket i mean that's that's pretty common i think in the message in them the way message passing is used so here as it's shown in figure 4.4 like although this is totally like beyond me to describe here this is a temporal message passing so it's like from one time point to another time point um and then on the left it depicts the hierarchical um like a hierarchical expansion collapsing over time steps but that a higher level network might predict the states and policies at the lower level and use these to draw inferences about the context in which these occur so the way that i've always seen the message passing used is from one time point to another or from one layer of a hierarchy to the next layer of a hierarchy and i've not seen it passed within constituents under the same markov blanket so eric if you have references that depict like some kind of message passing in active inference that's not through time steps or through hierarchical levels i would love to see that because i have not seen it well i'll just look at go back to box 4.1 where they talk about the variational message passing and how you you get information and i would say within a blanket you've got various nodes and they they have this hierarchical parent child relations and in order to do inference about hey what's you know what's our belief in one of these parents you got to kind of go up and down and say well what are the children of that and what do the other parents think so that's all within within a single markov blanket the parents and children the hierarchical relation and the message passing happens to to circulate that information within the blanket that's how i i read and i also put a link here to a paper by champion at all in which uh they have actually explicitly uh defined uh the active inference or better say reformulated the active inference formalism according to uh variational message passing and uh if you just give me a second i can find uh exactly the place where they have stated this in plain english oh you can continue with the other questions if you like but i need a moment to check this paper sure yeah i'll check it out also and yeah thank you for that so like when i'm reading this variational message passing this involves messages from all constituents of the markov blanket of x so the message is coming from all the constituents in the blanket but it doesn't say where the message is going so it does make sense like that i guess it that it that it's an interchangeable like or within would be a better word than from like within all constituents of the markov blanket because i was reading it as like the message is coming from everything under that blanket which i guess is an incorrect

interpretation um so thank you eric for pointing that out because i was like what so this is um a really hard question that i looked to try to answer um actually a bunch of us looked yesterday a lot so it says in equation 4.10 the sigma variable squiggle sigma we'll call it whatever i'm not sure how to say it is used to describe the difference between the natural log of observations conditioned on policy and preferences what is the function of this variable it also comes up in figure 4.4 the message passing figure we're just looking at and it would be great to have a verbal description of this figure what other papers or equations use this squiggle sigma and uh we looked at length like through the entire textbook through um the other chapters it refers to like we'll unpack this later in chapter seven we looked through chapter seven we looked through the appendices we saw a variable that almost looks like that but i think it's a gamma it looks like a little bit more fancy than this squiggle sigma and could not find at all so we went to unpack equation and so here is the equation itself i'm not sure how to put this up next to the actually maybe i'll put up here 419. so and this equation 4.10 a rewriting of equation 4.7 in linear algebraic form we tried to to kind of unpack this a little bit so the first line states the prior probability for each policy this is that π_0 is equal to the softmax function σ times the negative expected free energy g um and so that's this first line the next line is the expected free energy conditioned on policy g_{π} is equal to the entropy or negative expected log probability h times the states conditioned on policy and time which is s_{π} times τ or both of those s_{π} and τ plus the observations conditioned on policy and time this o_{π} times τ times the beliefs conditioned on policy and time which is the squiggle sigma maybe s_{π} times τ so so we were kind of unsure if beliefs is correct here we kind of pulled that out of the um legend for figure 4.4 but if anybody um knows and and just to maybe unpack that a little bit more the beliefs conditioned on policy and time the squiggle sigma s_{π} times τ is equal to the difference between the observations conditioned on policy and time o_{π} times τ and preferences conditioned on time c_{π} or c_{τ} so we were unsure if belief is correct here also and so if anybody knows if beliefs is um the difference between observations and preferences that would be great to have some kind of feedback or input here because we couldn't really reach a conclusion well i just i guess one one tiny step uh toward figuring this out i would say is that um that funny squiggle thing there has to do with the c we've got uh we've got this probability of observations given c and c is this mysterious object that expresses preferences which has also been underexplained that's so if you look at the two terms of expected free energy g as a function of a function of policy and 4.7 and then look at how it looks like in 4.10 that sigma thing there is it is expressed as is we've got the dot which i

guess is a multiplier by observation so that's the same as that's like saying your probability of observation given c so that's i guess a major matrix multiplied version of the um this log probability of um sequence of observations given your preference prior c yeah so they do unpack it a little bit more in equation 4.7 or it seems to be like maybe we could extrapolate from 4.7 if this is beliefs because it is this natural log of observations minus the natural log of preferences but but up in the top that just the free energy is um the entropy or the maybe the dot product of the entropy times the states plus the observations times or the dot product of the observations and maybe beliefs here um so i've not seen like i can't recall a mathematical representation of belief ever using this squiggle sigma before and i looked through um the recent ryan smith paper and i looked even at like the message passing paper by fristen and i was not able to really get any additional like references to this squiggle sigma at all so why do you say belief as opposed to preferences well so it says here that um the squiggle sigma natural log of the observations conditioned on policy and time and the natural log of the preferences at at a certain time so the sigma is defined by the natural log of the preferences but it's more than just the preferences does that make sense eric but that's not same as belief right that's that's observations versus right so the difference like rate from expected deviation or how much risk is being taken relative to the policy that's applied so because i i mean it's the dot product is between the observation and that's wiggle sigma right so which is in turn okay this is what i've observed this is this is what i expected so it's like a divergence between the two so how much more risk am i taking and how far is just pushing me away from my cream from minimizing free energy so maybe it's something like that whereas belief is wrapped up in the s because that's that's your model you know of the world so risk is maybe the how the the squiggle sigma is defined but then if you look into this figure 4.4 like that's that's why we used belief so here let's let's i'll pull it up right now so i'm not sure how easy this is to see but um in the before we get to number one here we have like at the second on the right hand side the second to lowest level is um states conditioned on policy and time i think i'm going to read it off this bigger screen oh yeah states conditioned on policy and time at different time steps so like the the current time is in the middle the forward time step is on the right and the backward time step is on the left um and so that's what we start with other times maybe beliefs is this s well so and that's kind of what what eric was saying um and we know because it's defined earlier this epsilon here is a prediction error right so so what we come to after step one it says and current outcomes to calculate prediction errors that's what we arrive at after number one um the errors then drive updating in the beliefs that's number two so here we go from the

current state at the present time to a prediction error that drives the belief updating at this forward time step after number two um and then it says given beliefs about states under each policy we can then calculate the gradients of expected free energy three so so what gives us this gradient of expected expected free energy is it s sub type so s sub τ plus one or is that a τ what i think that's a τ let's see so is it a future state not conditioned on policy or here we get to this squiggle σ so it's a squiggle σ conditioned on policy and time plus one so it looks like a belief update there which is why belief made sense so is that a risk update is that a possible interpretation there in this message passing figure well they say it's a gradient which means it's saying how much do we have to change our policy i think in order to get uh our objectives yeah gradient makes sense to me as well because uh it's how much information you've gained right so if you actually look at the change in the state with respect to so i apply some let's say force or like let's just take a simple example like if you have an actuator that just the applies force all it does is move moves forward and has to follow a trajectory the error is so at each time step if it's a deterministic system you expect it to follow a straight line and it's deviating up away so that's ϵ so you have to update okay this is how far up i am from the state that i'm supposed to be so i need to apply slightly less force or maybe a force in a different direction that seems so it's also how much i have to over correct in the future depending on what i've done currently right it's not just if i've applied some force right now i might deviate in the opposite direction and i have to come back so i have to figure out how much force to apply so that's why i said risk or yeah gradient you would have to take the uh change with respect to the state change with respect to the error so yeah that would be one interpretation of it right so it's a gradient of the state change with respect to what you have done so i also heard the term information gain in there um yeah yeah i mean i i mean i'm just generally saying so in neural networks for example we would have like some sort of loss function and we would calculate the weight change with respect to the change of the loss function right so like i have a weight vector and then uh this particular example is giving me a certain amount of gradient and i need to minimize this gradient so i need to change the weights in turn so the gradient actually interfaces between this loss function and all these weights so effectively the gradient passes on information as to how much the weight has to change in order to minimize that loss function that's why i said and also as a side note i think i found σ in ryan smith's paper too uh and it is defined as the expected prediction error the expected prediction error versus the actual prediction error like and and they distinguish it from ϵ in the ryan smith paper because i i did look at that but i didn't find it in there yeah

it's in question in equation 27 under the section outcome prediction errors it may be that the expected means it's an expectation over the q distribution which is your distribution over belief states yeah that's great ali like we were um we ali and i and um some others worked on this yesterday a lot and uh it's great that it kept you up late at night clearly ellie so you went digging around for for some more information that's awesome yeah great okay perfect well that's that resolves that very well and then let's get into maybe the next all right so in equation 4.16 the x has a dot over it changed through time but not the y data so there's some um discourse on this or maybe we should go to the next question actually because it's maybe more related so figure 4.6 uses an epsilon for prediction error as described in equation 4.21 is this predictive processing framing a part of the active inference model or is this presented for contrast to illustrate the simile similarities and differences between active inference and predictive processing so this is equation 4.1 and this is the epsilon but i think we see it way before equation 4.21 like it's even there in equation 4.4 um it says this predictive coding schema is part of the active inference model illustrating the hierarchical structure of predictions and beliefs the authors say one way to think about this is as if we had equipped a predictive coding scheme with classical reflex arcs at the lowest level of the hierarchy and they give this reference in this setting active inference is just predictive coding plus reflex arcs does anybody know what a reflex arc is because i'm a little bit lost with respect to that or have any comments on this prediction yeah reflex reflex arc is uh well basically the stimuli that doesn't go through the sensory cortex i thought it was like a mathematical construct maybe you think of it as a control system no thinking it's just it's like you know you bang your knee your foot goes up yeah that makes sense and it says we minimize free energy through action and the only part of the free energy that depends on action is the lowest level of prediction error in this hierarchical schema action fulfills descending predictions by minimizing the error between the predicted and observed sensory data any additional comments here about predictive coding and active inference or we can look at this equation 4.16 the x has the dot over it change through time but not the y data why is this and here the discourse says the top equation depends on $f(x)$ and v which is a deterministic function describing how a hidden state changes over time the bottom equation depends on $g(x, v)$ which describes how data are generated from a single hidden state the tilde indicates change through time the dot indicates the first order derivative any uh comments here on this equation or your thank you so maybe we can ask one more question or maybe a couple more so here it says on page 63 it says specifically the bayesian brain helps us frame the problems that an

agent engaging in active inference must solve broadly these are the problem of inferring states of the world perception and inferring a course of action planning other than perception and action planning are there other tasks or challenges that the brain or organisms engage in how would we know there's no discourse here but let's open it up and see what you guys think um yeah um people uh wonder hey uh what's on that star out there that's neither i don't know is that is that inferring state of the world i guess not planning and it could just be a hallucination of generative model you're not really measuring anything you're just measuring yourself over and over again that could be one thing okay so here there's another question for each of the graphical models in figure 4.2 what is an intuitive example for each structure one would be disease diagnosis right so you have symptom that is causative of or predictive of some condition i think that was the example that they gave in the text right or did we talk about this last week maybe yeah i actually don't get the question because reach of them we have an example at least an example in the text so yeah that's true and i think we're going to maybe see more examples as we go through the book but just in case if anyone wants to see some more examples uh actually that paper i put the link in the chat realizing active inference and variation message passing contains some more examples for each of these diagrams this is a fun last question maybe it says this reiterates that active inference uses two constructs variational free energy and expected free energy which are mathematically related but play distinct and complementary roles in your own words what are the definitions of variable variational free energy and expected free energy and what role do they play individually or together in active inference uh my view of expected free energy was uh given a certain action that might be taken it might produce this or reduce the free energy or something or it talks about some future uh reduction on the free energy or variational free energy is more objective i don't know how to put it so it's like it's like okay actual kinetic energy versus perceived kinetic energy kind of thing so you're rolling down a hill uh yeah like a ball rolling down a hill we can actually measure the kinetic energy at the bottom of the hill right if you put the ball at the top expected free energy would be like using a model to calculate that and then finding out it actually corresponds if you look at the model at least that's my view so i don't know and probably like i think about this in a lot of different ways but i think about like variational free energy i mean this is one way to think about it is like um maximizing like the the trade-off between epistemic value and pragmatic value like right now and expected for free energy is like maximizing that difference at some point in the future so you can like plan ahead like it might be cold out so i'm going to take a jacket or like it's cold right now i'm going to put my

jacket on those are kind of like that's how i think about the difference between those so it's ten thanks everyone for coming i hope i didn't um you know muddle this up too much uh trying to be a substitute teacher for daniel um and uh it's ten so we're gonna have tools right now in this room but if anybody wants to continue discussing these ideas and gather you're welcome to migrate up to one of the different spaces yeah and i think we'll stop the recording now alex if you haven't already